FastSpeech 2 Based Text-to-Speech Technical Report

Executive summary

The assignment asks for separate FastSpeech 2 text-to-speech models for English and Telugu, trained with approximately five hours of training audio and one hour of validation audio per language. The available evidence comes from a Colab T4 session and a local GitHub checkout. Telugu data selection and feature preparation were reported complete in Colab, and a short training run reached step 500 of 5,000. The runtime reset before any Telugu checkpoint was saved. English checkpoint files for steps 1,000 through 5,000 were observed in Drive, but the corresponding logs and samples are not present in this repository.

This is a progress report, not a final model evaluation. No validation-loss values, generated assignment test audios, per-sentence variance plots, or listening-quality observations are available for inclusion.

Workstream	Evidence available	Status
Reference source	FastSpeech 2 source and example configs in English/	Included
English run	Drive inventory showed checkpoints at 1k, 2k, 3k, 4k, 5k	Not independently evaluated
Telugu data	1,045 clips; 5.0005 h train and 1.0001 h validation	Prepared in Colab
Telugu training	Loss log through step 500 / 5,000; no saved checkpoint	Incomplete
Final inference/report assets	No test audio, variance plots, or validation curves found	Outstanding

2. Model and implementation approach

The project uses the public ming024/FastSpeech2 PyTorch implementation as its starting point. FastSpeech 2 is a non-autoregressive model that predicts a mel spectrogram from text. Its Transformer encoder produces text representations; a variance adaptor predicts duration, pitch (F0), and energy and expands the sequence; a Transformer decoder and postnet produce the final mel representation; a vocoder converts the mel spectrogram to audio. The reference implementation and paper are listed in References.

The local English/ directory contains the reference implementation and example configurations. The executed Colab work required compatibility changes for the installed librosa API and Telugu-specific text/data handling. Those runtime edits were not copied into this checkout, so their final source diff is not reproducible from the current repository.

3. Dataset selection and preprocessing

The assignment sheet names Svarah as the English dataset and IndicTTS Telugu for Telugu. The local source directory still contains upstream LJSpeech, AISHELL3, and LibriTTS example configurations. No Svarah selection manifest or English preprocessing log was found in the checkout; the observed English Drive checkpoints alone do not verify the English dataset or split.

3.1 Telugu data

The Colab session used the SPRINGLab/IndicTTS_Telugu Hugging Face dataset. The dataset page shown in the session reported 8,576 rows and a total size of 6.44 GB. The selection script scanned 1,356 rows before meeting its duration targets and wrote a selected manifest in Drive. It used a seeded random split: 70% of eligible rows for training, 15% for validation, and the remainder skipped, while accumulating clips to the target hours.

Telugu subset	Observed value
Selected clips	1,045
Training duration	5.0005 hours
Validation duration	1.0001 hours
Preprocessing output	1,045 utterances; 873 training items and 172 validation items

The observed Telugu preprocessing configuration used 22,050 Hz audio, 80 mel channels, an STFT filter length of 1,024 samples, and a hop length of 256 samples. Pitch and energy were configured as phoneme-level features with normalization. These values were read from the Drive configuration during the Colab session; the Telugu configuration files themselves are not in this GitHub snapshot.

The dataset audio and derived feature arrays remain outside the repository. The original IIT Madras dataset license was not verified in the available project artifacts. Confirm the license and course requirements before reusing or redistributing the corpus.

3.2 English data and artifacts

The course assignment specifies Svarah. The English/ directory is an upstream code/configuration snapshot whose examples include LJSpeech, AISHELL3, and LibriTTS; these are not proof that the assignment's Svarah split was prepared. During the Colab session, Drive contained English checkpoint filenames for training steps 1,000, 2,000, 3,000, 4,000, and 5,000. The checkpoint files and associated English training/validation logs were not copied into this public repository, so their model quality and training dataset cannot be assessed here.

> **Data limitation: the report does not include raw audio, feature arrays, a Telugu checkpoint, or English loss logs. Dataset licensing and English Svarah compliance still require confirmation.**

4. Training setup and observed Telugu run

The Telugu training configuration inspected in Drive targeted 5,000 steps. It logged every 100 steps, validated every 500 steps, requested synthesis every 2,000 steps, and saved checkpoints every 1,000 steps. The batch size was 16 and the learning-rate warmup was 4,000 steps. Because the runtime stopped at step 500, no checkpoint interval was reached.

Step	Total	Mel	Mel postnet	Pitch	Energy	Duration
100	11.5143	5.2397	5.2416	0.2895	0.4329	0.3106
200	8.9120	4.2411	4.1975	0.0847	0.2447	0.1441
300	6.1552	3.0051	2.8629	0.0455	0.1385	0.1031
400	3.9739	1.9440	1.7766	0.0328	0.1474	0.0730
500	2.8914	1.2880	1.4253	0.0269	0.0996	0.0516

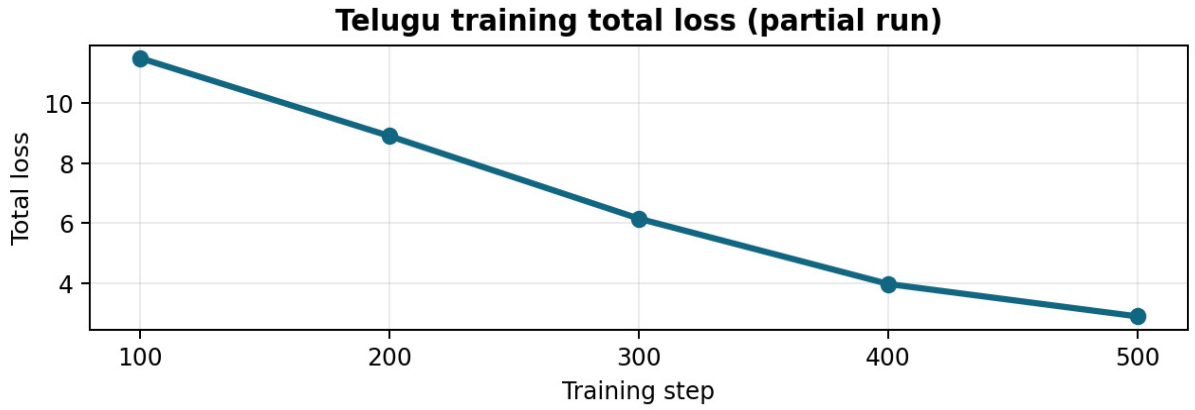


Figure 1. Telugu training total loss recorded at steps 100 through 500. This is a partial training curve, not a full convergence plot; no Telugu validation-loss series was available.

4.1 Loss observations

The logged total loss decreased at each observed 100-step interval, from 11.5143 to 2.8914 (about a 74.9% decrease over the logged range). Mel and mel-postnet losses account for most of the total. Pitch, energy, and duration losses were also lower at step 500 than at step 100, although energy increased slightly between steps 300 and 400. This pattern is consistent with early optimization progress, but five log points are not enough to establish convergence, stability, or generalization.

No validation values were captured in the available record. Without a validation curve and a saved checkpoint, overfitting, synthesis quality, pronunciation accuracy, and variance-adaptor behavior cannot be judged. The 500-step loss should not be presented as a completed training result.

5. Deliverable audit

Assignment item	Current repository status
English and Telugu preprocessing/training/inference code	English upstream code is included; complete Telugu code/config and executed notebook are not included.
Separate language configs and checkpoints	English checkpoint files were observed in Drive, not committed. No Telugu checkpoint was saved.
Training and validation loss plots	Only a chart of the partial Telugu training-loss observations is in this report. No validation or English curve is available.
Five test audios per language	Not generated or available in the repository.
Duration, F0, and energy plots	Not generated or available.
Technical report PDF	This partial report is provided; a final report requires the missing analyses and artifacts.
Public GitHub repository	Partial snapshot published at github.com/mishraayushgit/FastSpeech2-Assignment, branch main.

6. Limitations and next steps

The Colab runtime reset removed ephemeral Telugu data and features before a checkpoint was created. The local repository therefore preserves a progress record, not a resumable Telugu model. The executed notebook and the Colab-specific Telugu source patches are also absent from this checkout. The assignment's English dataset is Svarah, but the committed baseline includes upstream example configs and no Svarah split evidence. Raw data and weights were intentionally excluded from the public commit; the original Telugu dataset license was not verified.

To complete the assignment, restore the selected Telugu data/features from Drive or reproduce them; verify the Telugu license; save checkpoints at a shorter interval; complete training and validation; confirm that English uses the required Svarah subset; then generate the assignment's five sentences in both languages, save audio and duration/F0/energy plots, create loss curves for both splits, and revise this report with measured speech-quality observations.

7. Reproducibility

The project snapshot is at https://github.com/mishraayushgit/FastSpeech2-Assignment (branch main). The starter notebook in the repository is not the executed Colab notebook. The Colab session used a T4 GPU and reported Python 3.13, NumPy 2.4.6, librosa 0.11.0, and PyWorld 0.3.5. These are session observations, not a fully tested reproducible environment. Exact training commands, checkpoint files, full configs, and the executed Telugu preprocessing patch are not preserved in this snapshot.

References

[1] Y. Ren et al., *FastSpeech 2: Fast and High-Quality End-to-End Text to Speech*, arXiv:2006.04558, https://arxiv.org/abs/2006.04558
[2] ming024, *FastSpeech2 PyTorch implementation*, https://github.com/ming024/FastSpeech2
[3] SPRINGLab, *IndicTTS Telugu dataset*, https://huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu. Consult the dataset card and original license before use or redistribution.
[4] FastSpeech 2 assignment handout provided for this project, due 7 November 2026.

Status: partial progress only. This report deliberately does not infer missing validation, audio quality, or plot results.